

1. Executive Summary

This project implements a robust clothing segmentation system using DeepLabV3+ with a ResNet50 backbone, trained on the LIP (Look Into Person) dataset. The system achieves 84.83% mean Intersection over Union (mIoU) on clothes segmentation, demonstrating strong performance in separating clothing items from people and background. The model was trained using a novel combination of weighted Cross-Entropy and Dice loss to handle class imbalance and directly optimize for the IoU metric.

2. Dataset Choice and Reason

2.1 Dataset Selection: LIP (Look Into Person)

Dataset Overview:

- Name: LIP (Look Into Person)
- Size: 50,462 images with pixel-level annotations
- Annotations: 20 semantic parts including body parts, clothing items, and accessories
- Source: <https://www.kaggle.com/datasets/roneoz/lip-dataset/data>

2.2 Why LIP Dataset?

1. **Large-scale dataset**
The dataset contains more than 50,000 annotated images, providing significantly more training examples than a small custom dataset.
2. **Pixel-level annotations**
Since the task is semantic segmentation, pixel-level ground-truth masks are essential.
3. **Human-centric images**
The dataset focuses specifically on people, which matches the target application.
4. **Clothing-related semantic categories**
LIP provides separate labels for multiple clothing and human-part categories, allowing the original annotations to be transformed into a clothing-vs-non-clothing segmentation task.
5. **Challenging conditions**
LIP contains variations in pose, occlusion, viewpoint, and background complexity, which help the model learn more robust segmentation features.

2.3 Converting LIP into a Two-Class Problem

The original LIP dataset is a human-parsing dataset rather than a direct binary clothes-segmentation dataset.

Therefore, a label-mapping step was implemented.

The original semantic labels are mapped into:

New class	Meaning
0	Non-clothes
1	Clothes

This transformation allows the original multi-class human-parsing problem to be converted into the specific problem required by the project: **identifying clothing pixels rather than distinguishing every human body part.**

The preprocessing script performs dataset verification, checks the train/validation directory structure, validates split files, analyzes the mapped class distribution, and generates visualizations to verify the label mapping.

This step is particularly important because an incorrect label mapping would directly affect the quality of the training targets.

2.4 Image Preprocessing and Augmentation

All images are resized to **512 × 512 pixels** before being provided to the model.

Training augmentation includes:

- Horizontal flipping
- Random affine transformations
 - Scaling between 0.9 and 1.1
 - Translation
 - Rotation between -10° and +10°
- Color jitter
- Gaussian noise

These augmentations introduce controlled variations in the training data and help reduce overfitting to specific image appearances.

Validation images use resizing without the stochastic training augmentations, ensuring that evaluation measures the model on consistent inputs.

3. Model Architecture

3.1 DeepLabV3+

The selected architecture is **DeepLabV3+**.

DeepLabV3+ is an encoder-decoder semantic segmentation architecture introduced by Chen et al. The architecture combines multi-scale contextual information from Atrous Spatial Pyramid Pooling (ASPP) with a decoder that progressively recovers spatial information and improves object boundaries.

This is particularly relevant to clothing segmentation because clothing boundaries can be complex and may contain:

- thin structures,
- folds,
- overlapping garments,
- irregular boundaries,
- partial occlusions.

The decoder component is designed specifically to improve the spatial details of the segmentation output, especially around object boundaries.

3.2 ResNet-50 Encoder

The implementation uses **ResNet-50** as the encoder/backbone. The encoder is initialized using **ImageNet-pretrained weights**.

The purpose of the encoder is to extract increasingly high-level visual features from the input image.

Using pretrained weights provides two main advantages:

- The model starts with useful visual representations learned from a large image dataset.
- Training can converge more efficiently than starting from completely random weights.

The final segmentation head is configured for **two output classes** corresponding to non-clothes and clothes.

3.3 Architecture Overview

The overall pipeline can be summarized as:

Input Image (512 × 512 × 3)

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ResNet-50 Encoder

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DeepLabV3+ ASPP

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Decoder / Feature Refinement

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Segmentation Head

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2-Class Pixel Prediction

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Clothing Mask

The model is implemented using the `segmentation_models_pytorch` framework and configured through the project's YAML configuration file

4. Loss Function Selection

4.1 Combined Loss

The project uses a hybrid loss:

$$L = 0.5 * CE + 0.5 * Dice$$

where:

- CE = Cross-Entropy loss
- Dice = foreground Dice loss

The implementation explicitly combines these two components with equal weights.

4.2 Weighted Cross-Entropy

Cross-Entropy provides pixel-level classification supervision.

For each pixel, the model is penalized when it assigns a low probability to the correct class.

The project additionally uses class weights: [0.3, 0.7]

for the two classes:

- Non-clothes: 0.3
- Clothes: 0.7

The higher weight assigned to the clothing class encourages the model to pay more attention to the target foreground rather than simply optimizing for the dominant non-clothing region.

4.3 Dice Loss

Dice loss focuses directly on the overlap between the predicted segmentation and the ground-truth segmentation.

Dice-based objectives are particularly useful for segmentation because they directly optimize region overlap and can be more suitable than pixel-wise classification losses when foreground and background are imbalanced.

In this project, the custom Dice loss **ignores the background class** and focuses on the foreground clothing class. This is a deliberate design choice because the main objective is accurate clothing segmentation rather than accurate background prediction.

4.4 Why Combine Cross-Entropy and Dice?

The two losses provide complementary information.

Cross-Entropy:

- Provides strong pixel-level classification supervision.
- Encourages confident class predictions.
- Works naturally with the two-class classification formulation.

Dice Loss:

- Focuses on region overlap.
- Directly evaluates the quality of the clothing region.
- Reduces the tendency to obtain apparently good results simply by predicting the dominant background.

Therefore, combining them provides a balance between **pixel-level classification accuracy** and **foreground-region overlap**.

5. Performance Analysis and Evaluation

5.1 Evaluation Metrics

The evaluation system uses a **global confusion matrix** accumulated across the validation dataset. From this matrix, several segmentation metrics are calculated.

Intersection over Union

IoU measures the overlap between the predicted region and the ground-truth region.

The project reports:

- Per-class IoU
- Mean IoU (mIoU)

The mIoU is the average IoU across the two classes.

Pixel Accuracy

This measures the proportion of correctly classified pixels.

Precision

Precision indicates how many pixels predicted as clothing are actually clothing pixels.

Recall

Recall measures how much of the actual clothing region the model successfully detects.

F1 Score

The implementation reports macro precision, macro recall, macro F1, and per-class F1.

5.2 Final Results

Metric	Non-clothes	Clothes	Overall
IoU	0.895	0.808	0.852
F1	0.944	0.894	0.919
Pixel Accuracy	—	—	0.927

6. Limitations

Despite the use of a strong segmentation architecture and a large human-parsing dataset, the system has several limitations.

6.1 Binary Clothing Definition

The project converts the detailed LIP annotations into only two classes: **clothes** and **non-clothes**.

This simplifies the task but removes information about individual clothing categories.

For example, the model does not distinguish between:

- shirt,
- pants,
- skirt,
- dress,
- coat,
- etc.

6.2 Dataset Domain Gap

LIP contains human-centric images collected under varied conditions, but it may not perfectly represent images encountered in a real virtual fitting-room application.

For example, real users may provide:

- smartphone photographs,
- unusual camera angles,
- poor lighting,
- cluttered backgrounds,
- multiple people,
- partially visible bodies.

Therefore, performance on the LIP validation set may not directly represent performance on completely unseen real-world images.

6.3 Occlusion

Heavy occlusion remains a difficult segmentation scenario.

When clothing is partially hidden by:

- arms,
- other garments,
- objects,
- body parts,

the model may have insufficient visual information to determine the correct clothing boundary.

6.4 Fine Boundary Accuracy

Clothing boundaries can contain very fine structures such as sleeves, folds, straps, and edges.

Although DeepLabV3+ includes a decoder designed to improve object boundaries, segmentation at a fixed resolution of 512×512 can still lose very small details.

6.5 Computational Cost

ResNet-50 + DeepLabV3+ provides strong feature extraction capabilities, but it is heavier than lightweight segmentation architectures.

This creates a trade-off between:

segmentation quality ↔ inference speed ↔ memory consumption

For deployment on resource-constrained devices, a lighter encoder such as MobileNet could be considered.

6.6 Limited Experimental Comparison

The current project focuses on one main architecture and does not provide a comprehensive comparison against alternatives. Consequently, it is difficult to determine whether DeepLabV3+ is optimal for this particular dataset without additional controlled experiments.